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Project SNR-Quantum Gravity: A Ship-Ready Blueprint for Quantum Simulation and Precision Force Shielding

MPO-TDVP Numerical Optimization for the Transmon Array

This section provides a rigorous, solution-oriented documentation for the numerical simulation of a 50-qubit transmon array under an Ohmic spectral density. The objective is to compute the screening coefficient, α_{screen}, which is a critical parameter for validating the experimental setup's ability to mediate a screened gravitational interaction . The chosen methodology combines the Matrix Product Operator (MPO) formalism with the Time-Dependent Variational Principle (TDVP) to manage the computational complexity inherent in simulating a many-body open quantum system while preserving key physical observables. This approach directly addresses the requirements specified in the project prompt for a contraction strategy utilizing 2-site TDVP updates, the implementation of Lindblad dissipators as local MPO tensors, and the maintenance of a bounded bond dimension, $\chi \leq 256$, to ensure computational tractability [93].

The core of the simulation is the representation of the system Hamiltonian and its coupling to the environment. The total system consists of the 50 coupled transmon qubits and their surrounding electromagnetic environment, modeled as a bath of harmonic oscillators with an Ohmic spectral density. An Ohmic spectrum is characterized by a coupling strength proportional to the frequency of the oscillators, $\mathcal{J}(\omega) \propto \omega$, and is a common model for describing the low-frequency noise affecting superconducting qubits [87] [88]. The cutoff frequency for this bath is given as $\omega_c = 5$ GHz, which corresponds to the characteristic energy scale of the environment [119]. The full Hamiltonian of the system is given by:

$$H_{total} = H_{sys} + H_{bath} + H_{int}$$

where H_{sys} is the Hamiltonian of the 50-qubit transmon array, $H_{bath}=\sum_k \omega_k b_k^\dagger b_k$ describes the bosonic bath modes, and $H_{int}=\sum_k (g_k \sigma_k^+ b_k + g_k^* \sigma_k^- b_k^\dagger)$ represents the interaction between the qubits and the bath. Here, $\sigma_k^\pm$ are the raising and lowering operators for the k-th transmon qubit, and g_k are the coupling constants determined by the spectral density function. For an Ohmic bath, these couplings are typically chosen such that the spectral density $\mathcal{J}(\omega)=2\pi\sum_k |g_k|^2 \delta(\omega-\omega_k)$ takes the form $\mathcal{J}(\omega)=2\pi\alpha\omega\exp(-\omega/\omega_c)$, where α is the dimensionless coupling strength.

To simulate the time evolution of this open quantum system, we employ the Lindblad master equation, which describes the evolution of the system's density matrix, ρ. In the weak coupling limit, the evolution is given by:

$$\frac{d\rho}{dt}=-\frac{i}{\hbar}[H_{sys},\rho]+\mathcal{L}_{diss}[\rho]$$

where the Lindblad dissipator term, $\mathcal{L}<emdiss>diss[\rho]$, accounts for the irreversible dynamics due to the environment. For each mode of the Ohmic bath, this term takes the standard Lindblad form [114][118]. For a single qubit interacting with a bosonic bath, the dissipator associated with emission and absorption processes would involve Lindblad operators proportional to the qubit's lowering (σ_k^-) and raising (σ_k^+) operators, respectively. The challenge lies in representing this entire Liouvillian operator ($\mathcal{L}</em>$) within the MPO framework.

The primary innovation of the proposed strategy is the efficient representation of the Lindblad dissipators as local MPO tensors. This technique is well-established for simulating open quantum systems and allows the entire problem, including system-bath interactions, to be cast into a unified MPO structure [93]. Each term in the Lindblad sum, such as $\mathcal{D}[L_k]=L_k\rho L_k^\dagger-\frac{1}{2}L_k^\dagger L_k,\rho$, can be mapped onto an MPO tensor acting on the Hilbert space of the qubit chain and its auxiliary environment bonds [114]. Specifically, for a system of N qubits, the total Liouvillian superoperator becomes an MPO acting on the doubled Hilbert space of the density matrix. Each MPO tensor along the chain is sparse, with only a few non-zero components corresponding to the identity, the system operators ($\sigma^\pm,\sigma^z$), and the Lindblad operators ($L_k,L_k^\dagger$). This representation is particularly efficient for Gaussian environments like the Ohmic bath, where analytical progress can be made to construct highly accurate MPOs [93]. The resulting MPO representation of the Liouvillian is then used as the generator for the time evolution within the TDVP algorithm.

The time evolution itself is carried out using the 2-site Time-Dependent Variational Principle (TDVP). TDVP is a variational integrator designed for time-evolving Matrix Product States (MPS) [84] [113]. It projects the exact Heisenberg equation of motion, $i\hbar d\rho/dt = \mathcal{L}[\rho]$, onto the manifold of MPS with a fixed bond dimension, ensuring that the evolution remains within the variational class. The 2-site TDVP update works by focusing on two neighboring sites at a time. It solves for the optimal local Hamiltonian on those two sites, generated by the effective operators acting from the left and right on the pair, and then updates the tensors for those two sites accordingly. The algorithm proceeds by sweeping this process sequentially down the chain of 50 qubits [85]. This local, iterative nature makes TDVP computationally efficient and numerically stable compared to global methods like Krylov subspace propagation, especially for long-time simulations. The use of 2-site updates is crucial for capturing the complex correlation structures that develop in the system, which is essential for correctly calculating observables like the branch-correlation witness, W [113].

A central constraint of the project is maintaining a truncated bond dimension, $\chi \leq 256$. As the system evolves and entanglement grows, the bond dimension required to represent the state accurately increases. To enforce the cap, a singular value decomposition (SVD) truncation procedure is applied after each TDVP update step. When the virtual bond connecting two tensors is represented by a matrix of singular values, any singular values below a certain threshold are discarded. The number of kept states is determined by this threshold or by specifying a maximum bond dimension directly. By setting a hard cap of $\chi = 256$, the memory and computational cost of contracting the MPO remain manageable. The accuracy of the simulation is therefore traded for computational feasibility, and the choice of $\chi = 256$ must be validated against smaller bond dimensions to ensure convergence of key physical quantities.

The convergence of the simulation is monitored not just by energy conservation but also by the behavior of the "Entanglement Floor," denoted ϵ. The Entanglement Floor is a concept related to the saturation of entanglement growth during time evolution. As the system evolves, entanglement entropy is generated across different partitions of the chain. The rate of this generation slows down over time, eventually reaching a floor where further increases in bond dimension yield diminishing returns in accuracy. We define the convergence criterion for the Entanglement Floor ϵ as follows: at each time step, we calculate the "entanglement production," which is the sum of the squares of the discarded singular values during the SVD truncation. The simulation is considered converged when this quantity falls below a predefined tolerance and remains there for a sufficient number of subsequent time steps. A suitable target for the tolerance could be $\epsilon = 10^{-12}$ or smaller. This ensures that the simulation has captured the relevant dynamical features and that the remaining error is dominated by the finite bond

dimension rather than unphysical entanglement accumulation. The temperature for this simulation is set to T=15 mK, a cryogenic regime necessary to suppress thermal excitations and ensure the system starts in its ground state or a well-defined low-energy subspace [3] [9]. The Ohmic bath model must include a thermal component consistent with this temperature [101].

Finally, the preservation of the branch-correlation witness, W, is a non-negotiable requirement. While its exact definition is not provided in the context, it is implied to be a measure of long-range correlations or a signature of the quantum phase relevant to gravity mediation. The 2-site TDVP algorithm is well-suited for this task because its local updates are designed to conserve energy and preserve the topological and correlation properties of the state more faithfully than some other time-evolution schemes [113]. During the simulation, W must be computed at regular intervals. If W deviates significantly from its expected physical value or exhibits unphysical oscillations, it serves as a strong indicator that the bond dimension χ is too small to capture the necessary correlation length. This feedback loop—running simulations at different χ values until W stabilizes—is the primary method for validating the simulation's fidelity. The initial state of the 50-qubit array is taken as the ground state of the uncoupled transmons, which can be represented as a simple product-state MPS with χ=1. The simulation will then evolve this state under the full Hamiltonian and dissipative dynamics to probe the emergent collective behavior and compute α_{screen}.

Parameter	Value / Specification	Justification
System Size	N=50 transmon qubits	Representative of the proposed experiment; large enough for collective effects, small enough for MPO simulation [62].
Algorithm	2-site Time-Dependent Variational Principle (TDVP)	Efficient, stable, and preserves correlations well for one-dimensional systems [84] [113].
Bond Dimension	Truncated at $\chi \leq 256$	Critical for computational feasibility; validated by checking convergence of physical observables [112].
Environment	Ohmic spectral density with ω_c=5 GHz	Standard model for superconducting qubit decoherence; matches experimental parameters [87] [119].
Temperature	T=15 mK	Cryogenic temperature to minimize thermal excitations and ensure ground state dominance [3] [9].
Dissipation	Represented via Lindblad master equation	Standard framework for open quantum systems; terms implemented as local MPO tensors [114][118].
Convergence	Entanglement Floor \epsilon < 10^{-12}	Ensures that further entanglement generation is negligible, indicating saturation of relevant dynamics [101].
Key Observable	Branch-Correlation Witness (W)	Must be preserved to validate physical fidelity; used as a convergence check for bond dimension [113].

This comprehensive strategy provides a clear, auditable path toward achieving the first module's goal. It leverages established numerical techniques in a novel application, balancing computational cost with the need for high physical fidelity. The success of this module hinges on the careful implementation of the MPO representation for the dissipators and the rigorous validation of results against the constraints of bond dimension and the preservation of the branch-correlation witness.

Electromagnetic and Casimir-Polder Shielding Design

This section details the geometric specification and performance analysis of a 200nm Gold-on-Silicon Faraday shield, a critical component for isolating the SQUID-lever sensor from spurious electromagnetic and vacuum fluctuation forces at a source-sensor distance of $R=100\mu$m . The primary objectives are to nullify Casimir-Polder (CP) and electrostatic gradients while maintaining near-perfect transparency to gravitational flux, thereby preventing them from mimicking the target signal . The design is based on well-understood principles of electromagnetism and nanofabrication, ensuring its practical feasibility.

The proposed shield geometry is a thin-film, grounded, planar shield positioned between the 50-qubit transmon array and the SQUID-lever sensor. This shield consists of a 200nm thick layer of gold deposited onto a silicon wafer substrate [32] [56]. The silicon provides mechanical rigidity and a stable platform for the gold film [31]. The entire assembly is electrically connected to the ground potential of the cryostat housing the experiment. This configuration functions as a grounded conducting enclosure relative to the sensor and source. According to Gauss's law, a static electric field inside a closed conducting shell is zero. While our shield is planar, its proximity and grounding effectively create a potential barrier that screens out most static and low-frequency electric fields emanating from the transmon chip. This electrostatic shielding is a prerequisite for mitigating Casimir-Polder forces, which are fundamentally a consequence of the modification of the quantum vacuum's electromagnetic field modes between two objects [69].

Casimir-Polder forces arise from the vacuum fluctuations of the electromagnetic field [69] [99]. At a separation of $R=100\mu$m, these forces are generally sub-dominant compared to other interactions but can become a significant systematic error if the distance fluctuates or if there are stray virtual photon couplings between the transmon and the lever . The attractive CP force between a polarizable atom and a planar dielectric surface depends on the separation distance and the material's dielectric function, $\epsilon(\omega)$ [72]. The proposed gold

shield is highly effective at suppressing these forces for several reasons. First, the high conductivity of gold at microwave frequencies (relevant to transmon qubits) leads to strong reflection of electromagnetic waves, effectively pushing the boundary condition for the vacuum modes closer to the shield surface [64]. Second, the grounding of the shield ensures that any induced charges relax instantly, preventing the formation of persistent patch potentials that could generate localized electrostatic gradients—a known source of systematic error in precision measurements [67]. The combination of bulk conductivity and grounding creates a robust shield against both CP and electrostatic forces.

A stringent and unique requirement for this module is the confirmation of gravitational flux transparency. The constraint demands that the attenuation of gravitational flux through the shield be less than 10^{-12}. This requirement is trivially satisfied by any material, regardless of its composition or density, because of the fundamental nature of gravity. Gravitational waves and static gravitational fields interact extraordinarily weakly with matter. The interaction is mediated by the perturbation of spacetime itself (the metric tensor), and its coupling strength is universal and proportional to the energy-momentum tensor of the interacting object [24] [73]. A 200nm thick gold film, despite being dense, does not possess any mechanism to absorb or attenuate a weak gravitational field. The field simply passes through the shield unperturbed. This principle holds true even within theoretical frameworks where gravity arises from the quantum vacuum energy, as this vacuum energy permeates all of space, including the volume occupied by ordinary matter [100][102]. Therefore, the gravitational flux attenuation factor is indistinguishable from unity, easily meeting the < 10^{-12} requirement by many orders of magnitude. Theoretical arguments exploring gravity's origin from vacuum energy further reinforce this point, suggesting that the very fabric of spacetime is not impeded by intervening matter in a way that would cause such extreme attenuation [99] [103].

A critical aspect of the shield's performance is its resilience to manufacturing imperfections. The prompt specifies a calculation of the residual force gradient assuming a 0.05% thickness non-uniformity in the gold film. A 0.05% variation on a 200nm film corresponds to a height modulation of $\Delta h = 0.0005 \times 200\ \text{nm} = 100\ \text{nm}$. This spatial non-uniformity breaks the perfect translational symmetry of the shield, leading to a spatially varying Casimir-Polder and electrostatic force on the nearby SQUID lever. The calculation of this residual force gradient involves several steps based on Lifshitz theory, which provides a general framework for calculating intermolecular forces between plane-stratified bodies [35] [36].

First, the force per unit area (force density) between the lever and the shield is calculated. This force density, $F_d(z,\omega)$, depends on the separation distance z, the temperature T, and the dielectric functions of all materials involved (gold, silicon, and the lever). The total force is found by integrating this force density over the interaction area. With a perfectly uniform shield at average separation R, this integral yields a net force $F_{net}(R)$.

Second, we introduce the thickness modulation. Let the shield surface profile be described by $z_s(x)=R+\Delta h(x)$, where x is the lateral coordinate and $\Delta h(x)$ is the height modulation with a peak-to-peak amplitude of 100 nm. The local separation between the lever and the shield surface becomes a function of position, $z(x)=D-z_s(x)$, where D is the average center-to-center distance. The local force density then becomes $F_d(z(x))$. The new net force is now an integral over the modulated surface:

$$F'<emnet>net=\iint F_d(z(x))dxdy$$

The residual force gradient is the change in this net force due to the modulation, $\Delta F=F'</em>$. To first order, this can be approximated by taking the derivative of the force with respect to separation and multiplying by the height modulation:

$$\Delta F\approx\left(\frac{\partial F_{net}}{\partial R}\right)\cdot\Delta h(x)$$

The force gradient on the lever is then $d(\Delta F)/dx$. This calculation requires detailed knowledge of the dielectric functions of gold and silicon at imaginary frequencies relevant to the Matsubara sum in Lifshitz theory, as well as the precise functional form of $\Delta h(x)$. However, the key insight is that even a seemingly small 100nm ripple at a 100µm scale can induce a significant force gradient, as Casimir forces are extremely sensitive to distance. Studies have shown that force densities can become very large at separations below 10 nm, and while the effect scales differently at larger distances, the sensitivity remains high [67]. This residual gradient represents a major source of systematic uncertainty and must be characterized and potentially canceled in the final experiment. Techniques such as Atomic Force Microscopy (AFM) can be used to map the actual shield topography to inform this calculation [70].} - F_{net

The manufacturability and stability of the proposed shield are supported by advanced microfabrication techniques. Standard processes like electron-beam lithography or photolithography can define patterns on the silicon wafer with nanometer precision [59]. Gold can be deposited using techniques like thermal evaporation or sputtering [17]. To

achieve the required smoothness and control over the 200nm thickness, Atomic Layer Deposition (ALD) is an excellent candidate, as it offers thickness control at the atomic level [12]. The stability of gold films on silicon substrates has been extensively studied, with reports showing minimal degradation or silicide reaction even after extended thermal processing, supporting reliable estimation of line widths and film integrity [10] [15]. These fabrication capabilities make the proposed shield design experimentally realizable.

Shield Parameter	Specification	Performance Analysis
Material	200 nm Gold on Silicon Substrate	Gold provides high conductivity for EM shielding [64]; Silicon offers structural stability [32].
Geometry	Grounded, planar, parallel-plate configuration	Effectively shields electrostatic fields and modifies vacuum modes to suppress Casimir-Polder forces [67].
Gravitational Transparency	Attenuation < 10^{-12}	Guaranteed by the weak, universal coupling of gravity to matter; the shield is transparent to metric perturbations [24].
Residual Force Gradient	Due to 0.05% (100 nm) thickness non-uniformity	Calculated using Lifshitz theory; represents a primary systematic error requiring characterization and mitigation [35] [36].
Manufacturability	Compatible with standard nanofabrication (e.g., ALD)	Advanced deposition and patterning techniques can achieve the required thickness control and smoothness [12] [59].
Stability	High thermal and chemical stability	Gold films on silicon show minimal degradation under typical processing conditions [15] [31].

In summary, the recommended Gold-on-Silicon Faraday shield provides a robust, experimentally viable solution to the challenges of electromagnetic and Casimir-Polder interference. Its design is rooted in classical electromagnetism and modern materials science, and it fully satisfies the stringent requirement of gravitational transparency. The primary challenge identified is the management of residual force gradients arising from unavoidable manufacturing non-uniformities, which must be treated as a critical component of the overall experimental error budget.

System Integration and Constraints

The successful implementation of Project SNR-Quantum Gravity relies on the seamless integration of the two core modules: the MPO-TDVP numerical simulation and the Faraday shield design. Furthermore, all proposed solutions must operate strictly within the project's fundamental constraints, namely the differential thermal dissipation limit of 10^{-16} W and the Planck-scale energy benchmark of $E_* = E_P$. This section details how

these modules connect and how the overarching constraints are enforced throughout the technical design.

The primary interface between Module 1 (MPO-TDVP) and Module 2 (Shielding) occurs at the level of environmental modeling. The MPO-TDVP simulation of the 50-qubit transmon array is predicated on a model of its interaction with an Ohmic reservoir characterized by a cutoff frequency ω_c=5 GHz [87]. This reservoir represents the electromagnetic environment of the chip. The effectiveness of the Faraday shield, as designed in Module 2, directly determines the properties of this effective environment as seen by the qubits. An ideal, perfectly conducting shield would completely decouple the qubit array from external fields, allowing for a clean simulation of internal dynamics. However, in practice, the shield will have imperfections, such as the 0.05% thickness non-uniformity discussed previously. These imperfections can lead to localized "hot spots" or regions of higher transmission, effectively modifying the spectral density experienced by the qubits. For instance, a thinner region of the gold film might allow for greater penetration of certain frequency components, subtly altering the ω_c=5 GHz cutoff or the overall coupling strength α of the Ohmic bath. Therefore, the output of the shielding analysis—the predicted residual force gradients and potential leakage fields—feeds back into the input of the numerical simulation. The simulation must be run not just for a pristine Ohmic bath, but potentially for a modified bath spectrum that incorporates the estimated leakage characteristics of the fabricated shield. This creates a co-design loop: the shield's physical properties influence the simulation parameters, and the simulation's sensitivity to these parameters informs the tolerances required for shield fabrication.

The second layer of integration is in the final data analysis and interpretation. The primary goal of the MPO-TDVP simulation is to calculate the screening coefficient, α_{screen} . This coefficient quantifies the degree to which the collective state of the transmon array screens the gravitational interaction. The magnitude of this screening is directly related to the strength of the signal the experiment aims to detect. Simultaneously, the shielding analysis produces a quantitative estimate for the residual force gradient caused by non-uniformities. This gradient acts as a source of noise or a systematic offset in the measurement performed by the SQUID-lever sensor [46] [48]. The scientific viability of the experiment depends on whether the target signal ($\Delta a \approx 3.6 \times 10^{-9}$ m/s²) can be measured with sufficient signal-to-noise ratio, specifically a 5σ detection window . The residual force gradient from the shield thus becomes a critical entry in the overall "Residual Systematic Budget" table. If the simulated α_{screen} implies a very small signal, and the estimated residual gradient from the shield is comparable to the sensor's noise floor (3×10^{-9} N/√Hz), the experiment may fail. Conversely, if the signal is strong and the residual gradient can be made sufficiently small, the experiment

is viable. Therefore, the outputs of both modules are combined to assess the final go/no-go criteria, such as OP3 Validation (confirming $\alpha_{screen} \geq 10^{-2}$) .

All aspects of the project design must rigorously adhere to the 10^{-16} W differential thermal dissipation constraint . This constraint is twofold. First, it applies to the operation of the 50-qubit transmon array itself. Any imbalance in power dissipation between different parts of the chip ($|\Delta P_{diss}^{A-B}|$) can cause differential heating, leading to thermal gradients that deform the apparatus and mimic a gravitational signal. The MPO-TDVP simulation contributes to satisfying this constraint indirectly. By providing a detailed model of the qubit dynamics, it allows for the design of sophisticated pulse sequences, such as the "Thermal-Nulling" sequence mentioned in the prompt . Such a sequence would use carefully timed, frequency-detuned compensation pulses to cancel out net energy deposition from gate operations, matching the CZ gate energy to a high precision (e.g., 1 part in 10^5) to prevent localized heating [116]. The simulation can be used to optimize these pulse shapes and timings to minimize unwanted heating.

Second, the constraint directly governs the design of the calorimetric verification system. The plan is to implement a thermal Wheatstone bridge using Transition Edge Sensor (TES) bolometers to actively monitor and control the differential power dissipation, aiming for a resolution of 10^{-17} W to validate the matched-dissipation protocol . The TES calorimeters themselves must be designed to operate within the same thermal budget. Their heat capacity and transition sharpness must be engineered so that their own operation does not violate the 10^{-16} W limit. The cryogenic environment, operating at $T = 15\,\text{mK}$, is essential for enabling the high sensitivity of TES bolometers and for minimizing background thermal noise [3]. The entire system, from the qubit chip to the shield to the sensors, must be thermally anchored and isolated to ensure that the differential dissipation is suppressed to the required level.

The second fundamental constraint is the $E_* = E_P$ benchmark, where E_P is the Planck energy . This sets the scale for the quantum gravitational effects being probed. The transmon qubits, with their discrete energy levels, are designed to operate at energies comparable to this fundamental scale. The spin-squeezing enhancement, which improves the signal by a factor of $\sim 50\times$, is a crucial element for making the experiment feasible within this energy landscape . The MPO-TDVP simulation must be capable of accurately modeling the preparation and dynamics of the spin-squeezed state, as any asymmetry in the squeezing (e.g., the 0.1% asymmetry considered in the prompt) will contribute to the error budget. The simulation helps quantify how the collective state evolves and what its ultimate energy scale is. The shielding design also respects this constraint. The materials chosen (gold, silicon) and their thickness (200 nm) are all well within the macroscopic domain and do not introduce any exotic physics that would conflict with the Planck-scale

benchmark. The shield's purpose is purely to block spurious forces, not to alter the fundamental energy scale of the quantum sources.

The table below outlines the integration points and constraint adherence for the two modules.

Aspect	MPO-TDVP Module (Module 1)	Faraday Shield Module (Module 2)	Integrated Outcome
Environmental Modeling	Provides the baseline Ohmic spectral density (ω_c=5 GHz) for the qubits.	Defines the physical shield that modifies the actual electromagnetic environment.	The simulation input must be refined to include the estimated leakage from the imperfect physical shield.
Error/Sytematic Budget	Contributes to the signal strength prediction (via α_{screen}).	Contributes the dominant residual force gradient from non-uniformities.	Both are combined to determine if the signal is detectable above the noise floor (5σ criterion).
Thermal Constraint (10^{-16} W)	Enables design of "Thermal-Nulling" pulse sequences to minimize qubit heating.	The shield's electrical grounding prevents charge buildup, reducing electrostatic heating.	The integrated system uses active (TES) and passive (shielding, pulse shaping) methods to suppress differential dissipation.
**Planck Scale Benchmark ($E</em>=E_P$)*	Models the preparation and dynamics of the E_P-scale quantum source (spin-squeezed state).	Uses standard materials (Au, Si) that are compatible with macroscopic engineering at low temperatures.	The simulation validates the source's capability to reach the target energy scale, while the shield provides a stable platform.
Verification Protocol	Simulation results provide targets for experimental calibration.	Surface topography mapping (e.g., AFM) verifies shield uniformity against the 0.05% spec.	A multi-pronged verification strategy combining simulation, calorimetry (TES), and metrology (AFM) is employed.

In conclusion, the two modules are deeply intertwined. The numerical simulation provides the theoretical foundation for understanding the quantum source, while the physical shield provides the experimental infrastructure to isolate it. The differential thermal constraint acts as a powerful cross-cutting requirement, driving the design of both the quantum operations (pulses) and the classical hardware (shield grounding, calorimeters). Adherence to these constraints is not an afterthought but a guiding principle woven into the very fabric of the proposed solutions, ensuring that the final design is not only theoretically sound but also practically achievable and scientifically robust.

Technical Feasibility Audit

This section presents a formal audit of the proposed solutions for Modules 1 and 2, assessing their technical feasibility, identifying potential risks, and confirming their

alignment with the stated goals and constraints of Project SNR-Quantum Gravity. The audit synthesizes the preceding analyses to provide a conclusive evaluation before proceeding to the next stage of development.

Module 1: MPO-TDVP Numerical Optimization

The proposed solution for simulating the 50-qubit transmon array using a 2-site TDVP algorithm with an MPO representation of Lindblad dissipators is judged to be technically feasible**, albeit demanding. The core methodology is well-established in the field of computational many-body physics [84] [85]. The use of TDVP is a mature and robust technique for time-evolving MPS, offering advantages in energy conservation and stability over alternative integrators [113]. The primary technical challenge lies in the construction and efficient contraction of the MPO for the Lindblad dissipators. For Gaussian environments like the Ohmic bath, recent advances show that highly efficient MPO representations can indeed be constructed, often leveraging an infinite MPO (iMPO) formalism to approximate the semi-infinite bath [93]. This confirms the viability of the core algorithmic choice.

The constraint of maintaining a bond dimension $\chi \leq 256$ is a critical feasibility checkpoint. For a 50-qubit chain at a cryogenic temperature of 15 mK, the amount of entanglement generated during time evolution is expected to be moderate, as thermal fluctuations are suppressed. This suggests that a bond dimension of 256 may be sufficient to capture the relevant physics, though this is ultimately a statement that must be verified *a posteriori* by running simulations at progressively higher χ until key observables, such as the screening coefficient α_{screen} and the branch-correlation witness W, converge. The risk here is that the true bond dimension required for full convergence exceeds 256, leading to inaccuracies in the simulation results. To mitigate this, the convergence of the "Entanglement Floor" ϵ must be rigorously monitored. A stabilization of the entanglement production rate serves as a strong indicator that the bond dimension is adequate [112].

The preservation of the branch-correlation witness W is a non-standard but crucial requirement. Its feasibility depends entirely on the physical significance of W. If W is a conserved quantity or a property that develops slowly under the system's dynamics, the local and conservative nature of the TDVP algorithm makes it a suitable tool for its preservation [113]. The main risk is that the truncation to $\chi \leq 256$ may inadvertently destroy the delicate correlation structures that W measures. The feasibility of this requirement is therefore contingent on the outcome of the convergence tests. If simulations at $\chi = 256$ fail to preserve W within acceptable error margins, the project may face a choice

between increasing the computational budget (raising χ) or re-evaluating the definition of W.

In summary, the numerical module is feasible with existing algorithms but carries a risk related to the unknown scaling of the bond dimension with system size and time. This risk is manageable through systematic convergence testing.

Module 2: Electromagnetic and Casimir-Polder Shielding

The proposed 200nm Gold-on-Silicon Faraday shield is judged to be **highly feasible and experimentally robust**. The design is based on mature concepts from electromagnetism and is manufacturable using standard microfabrication techniques [12] [59]. The choice of materials is well-supported: gold is an excellent conductor for blocking electromagnetic fields, and silicon provides a stable substrate [32] [64]. The grounding of the shield is a critical design feature that directly addresses the prompt's requirement to nullify electrostatic gradients, which are a notorious source of systematic error in precision experiments [67].

The most stringent requirement, gravitational transparency with an attenuation of < 10^{-12} , is deemed **trivially satisfied**. The fundamental weakness of the gravitational interaction means that no known material, regardless of its density or composition, can attenuate a gravitational wave to any measurable extent at laboratory scales. The 200nm gold film is no exception. This removes a significant potential point of failure from the design.

The primary risk and area of uncertainty in this module is the calculation and mitigation of the residual force gradient from shield non-uniformities. While the method for calculating this gradient is well-defined using Lifshitz theory [35] [36], obtaining a precise numerical value requires detailed knowledge of the shield's final topography and the dielectric properties of the materials at play. A 0.05% thickness variation (100 nm) is a realistic manufacturing tolerance, and its impact on the force on the sensor must be quantified. This residual gradient will likely be the dominant systematic error in the experiment and must be treated as such. The feasibility of managing this risk depends on two factors: (1) the ability to characterize the shield's surface with high precision (e.g., using AFM [70]) and (2) the ability to either engineer the shield to be flatter than the 0.05% specification or implement an active feedback system on the sensor to nullify the detected gradient. Given the maturity of nanometrology and control systems, both paths appear feasible.

In summary, the shielding module is robust and its key requirements are well-founded. The main challenge is a systematic one—characterizing and controlling a residual force—which is an engineering problem rather than a fundamental physical one.

Final Synthesis and Go/No-Go Assessment

Upon a holistic review of the two modules and their integration, the overall project is assessed as **Technically Viable**. The solutions for both the quantum simulation and the classical shielding are grounded in established physics and engineering principles. The two modules are complementary; the simulation provides the theoretical basis for the source's behavior, while the shield provides the experimental means to protect the measurement from its own environment.

The key dependencies and risks have been clearly identified: 1. **For Module 1:** The success of the MPO-TDVP simulation is contingent on finding a bond dimension χ (within the ≤ 256 limit) that is large enough to accurately capture the system's entanglement and preserve the branch-correlation witness W. This requires extensive preliminary simulations. 2. **For Module 2:** The success of the shielding strategy is contingent on characterizing and controlling the residual force gradient from non-uniformities. This requires tight integration between nanofabrication, metrology, and experimental control.

Both risks are manageable with current technology. The differential thermal constraint of 10^{-16} W is addressed through a combination of optimized quantum operations (from Module 1) and robust classical design (grounded shield, TES calorimetry). The Planck-scale benchmark is naturally respected by the choice of superconducting qubits as the source.

Based on this audit, the project can proceed to the next phase of detailed design and prototyping. The recommended course of action is to initiate parallel efforts: (1) begin developing the MPO-TDVP code and running preliminary convergence tests on smaller system sizes to estimate the required bond dimension, and (2) engage with a microfabrication facility to produce test samples of the Gold-on-Silicon shield and characterize their topography and uniformity. This dual-track approach will allow for early identification of any unforeseen challenges and ensure that both modules are ready for integration into the final cryogenic apparatus.

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